TraceScope AI 2.0: Dual-Branch Deep and Handcrafted Forensic Fusion for Explainable, Leakage-Free Flatbed Scanner Attribution, Open-Set Device Rejection, and Tampering Localization

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Executive Abstract

Attributing questioned digital documents to specific flatbed scanner hardware is a critical challenge in forensic informatics, legal chain of custody validation, and anti-counterfeiting. Traditional sensor pattern noise methods based on Photo-Response Non-Uniformity (PRNU) degrade substantially on text documents due to high-contrast typography, while standard deep convolutional neural networks (CNNs) are prone to "shortcut learning" (memorizing printed characters rather than hardware sensor artifacts) and suffer from cross-split document patch leakage.

In this paper, we introduce **TraceScope AI 2.0**, an end-to-end, scientifically audited forensic framework for leak-free scanner attribution, open-set rogue scanner rejection, and localized tampering detection. We construct a strictly partitioned, document-level dataset comprising **4,568 high-resolution document scans** spanning **11 flatbed scanner units** across 3 manufacturers (Canon, Epson, HP) at 300, 400, 600, and 1200 DPI, with **0.00% cross-split leakage**.

We establish a comprehensive 3-tier comparative benchmark:

1. **Tier 1 (Classical ML Baselines):** Random Forest achieves **58.53%** test accuracy (Macro-F1: 0.5787), while Support Vector Machines (RBF) achieve **33.97%** accuracy (Macro-F1: 0.3346) on 10 statistical noise moments.
2. **Tier 2 (Deep CNN Baseline):** A PyTorch ResNet-18 operating on Kraetzer-Vogler high-pass filtered residuals achieves **97.35%** closed-set test accuracy (Macro-F1: 0.9735).
3. **Tier 3 (Flagship Dual-Branch Hybrid CNN):** A custom neural architecture fusing a spatial residual CNN branch with a 44-dimensional handcrafted descriptor branch (11 PRNU moments, 3 2D-FFT radial spectral energy ratios, 26 LBP bins, and 4 GLCM Haralick moments), achieving **82.35%** test accuracy (Macro-F1: 0.8231).

A rigorous 6-configuration systematic ablation study demonstrates that dual-branch late fusion (**82.35%**) delivers a **+18.53%** gain over the isolated CNN branch (**63.82%**) and a **+6.61%** gain over handcrafted features alone (**75.74%**), confirming strong physical-deep feature synergy. Under 29 operational degradation evaluations (7 transformation families), the system exhibits near-perfect contrast invariance (). For open-set forensic deployment, while standard Softmax Maximum Probability fails catastrophically due to extreme out-of-distribution overconfidence (AUROC 1.19%), **Penultimate Latent Distance Rejection** achieves **98.57% AUROC** (88.29% AUPR, 3.06% FPR@95%TPR). Furthermore, patch-level residual

variance ratio mapping localizes digital inpainting/erasure with **58.78% precision** and **0.00% false alarms** on authentic scans, while Gradient-weighted Class Activation Mapping (Grad-CAM) quantitatively proves typographic decoupling (). Finally, we demonstrate full integration with an interactive web dashboard and a Federal Rule of Evidence (FRE) 902(14) compliant two-page cryptographic forensic dossier exporter.

Index Terms: Digital Document Forensics, Flatbed Scanner Attribution, Photo-Response Non-Uniformity (PRNU), Dual-Branch Neural Networks, Open-Set Scanner Rejection, Document Inpainting Localization, Explainable AI (Grad-CAM), Daubert Standard Admissibility.

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1. Introduction & Forensic Problem Formulation

Digital flatbed scanners serve as the primary bridge converting physical evidentiary media - wills, financial contracts, real estate titles, bank cheques, and official identity documents - into digital bitstreams. In forensic casework and questioned document examination (QDE), investigators frequently face two foundational questions:

1. **Device Attribution:** *Was Document X digitized on Scanner Y owned by the suspect, or on an unrelated machine?*
2. **Document Integrity & Tampering:** *Has any portion of Document X been digitally altered, erased, or inpainted subsequent to scanning?*

1.1 Physical Sensing & Substrate Noise Obstacles

In digital camera forensics, Photo-Response Non-Uniformity (PRNU) - arising from sub-micron silicon manufacturing defects in pixel sensor wells - provides a permanent, unique hardware fingerprint. However, applying PRNU modeling to flatbed document scanners introduces severe physical and mathematical obstacles:

- **Spatial Sparsity of Sensor Noise:** Text documents predominantly consist of saturated black ink text over high-reflectance white paper backgrounds, drastically starving high-pass filters of mid-tone continuous sensor noise.
- **Substrate Fiber & Halftoning Interference:** Paper cellulose fibers, wood pulp irregularities, and print halftoning generate spatial textures that completely overwhelm the microscopic PRNU signal ().
- **Mechanical Scan-Line Jitter:** Unlike 2D staring array camera sensors, flatbed scanners acquire documents via a 1D linear CCD/CIS array propelled mechanically across the platen by a stepper motor and belt, introducing non-uniform velocity fluctuations and scan-line synchronization drift.

1.2 The "Shortcut Learning" Crisis & Document-Overlap Leakage

Recent academic attempts to classify scanner brands using Deep Convolutional Neural Networks (CNNs) frequently report near-100% classification accuracy. However, thorough forensic audits reveal that these high scores are predominantly artifacts of **methodological flaws**:

1. **Shortcut Learning on Font/Paper Artifacts:** Deep CNNs optimize purely for empirical risk minimization. When presented with unfiltered document patches, the network learns to identify specific printer halftoning patterns, microscopic font edge kerning, or document layout geometries, completely ignoring the scanner hardware. Once tested on a different document printed with a different printer, performance collapses to random chance.
2. **Patch-Level Splitting Leakage:** Prior studies routinely slice scanned document pages into hundreds

of sub-patches and randomly shuffle them into training and testing sets. Because patches from the same physical document share the exact same paper grain, typography, ink darkness, and acquisition lighting, the network simply memorizes page identities rather than scanner hardware.

1.3 Key Scientific Contributions of TraceScope AI 2.0

To overcome these limitations and build a legally defensible forensic framework, TraceScope AI 2.0 introduces the following contributions:

- **Leak-Free Document Partitioning:** We construct a strictly audited benchmark of **4,568 document images** from **11 physical scanner devices**, partitioned strictly at the physical document level (0.00% cross-split leakage verified via SHA-256 manifests).
- **Tri-Tier Empirical Benchmark:** We implement and compare classical statistical ML models (Random Forest, SVM), deep convolutional models (ResNet-18 with Kraetzer-Vogler filtering), and our proposed **Flagship Dual-Branch Hybrid CNN**.
- **Dual-Branch Fusion Architecture:** We combine a spatial residual CNN branch with a 44-dimensional handcrafted descriptor branch (11 PRNU moments, 3 2D-FFT radial energies, 26 LBP bins, and 4 GLCM Haralick moments), achieving **82.35%** test accuracy and proving +18.53% synergy over isolated CNNs.
- **Penultimate Latent Space Open-Set Rejection:** We expose the complete breakdown of Softmax overconfidence on out-of-distribution devices (AUROC 1.19%) and engineer a penultimate Euclidean centroid distance metric that achieves **98.57% AUROC** (88.29% AUPR, 3.06% FPR@95%TPR).
- **Localized Anomaly Detection for Tampering:** We develop a patch-level residual variance ratio engine that localizes digital text erasure/inpainting with **58.78% precision** and **0.00% false alarms** on authentic scans.
- **Anti-Shortcut Explainability Audit:** Using Grad-CAM, we prove quantitatively that convolutional activation heatmaps decouple from macroscopic text edges (), confirming true forensic feature learning.
- **Court-Admissible Dossier Export:** We integrate the forensic engine with an automated reporting pipeline generating cryptographic, FRE 902(14) compliant PDF evidence dossiers.

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2. Theoretical Principles & Mathematical Formulations

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                    [ Input Document Image I(x,y) ]
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2.1 Optical Sensor PRNU Model

A flatbed scanner sensor acquires optical radiance according to the linearized sensor model:

where is the digitized pixel intensity, represents the ideal optical document scene, is the zero-mean multiplicative Photo-Response Non-Uniformity (PRNU) matrix characteristic of the individual sensor array, and is additive zero-mean Gaussian read noise.

2.2 Kraetzer-Vogler High-Pass Residual Filtering

To isolate and remove the dominant macroscopic text edges , we apply an adaptive spatial unsharp Kraetzer-Vogler filter with a Laplacian operator :

The residual array is computed as:

Residuals are DC-centered () and standardized to zero-mean unit-variance () across spatial patches.

2.3 44-Dimensional Handcrafted Feature Formulation

In parallel with deep spatial convolutions, each document patch is transformed into a 44-dimensional deterministic feature vector :

1. **PRNU Cross-Correlation & Higher-Order Moments (11 Dimensions):**

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   - Peak-to-Correlation Energy (PCE) and Normalized Cross-Correlation (NCC) against pre-computed manufacturer reference flatfield matrices:
   - Statistical moments of the noise residual distribution:
   - Shannon multi-scale entropy of the noise histogram.
2. **2D-FFT Radial Spectral Energy Ratios (3 Dimensions):**
   The 2D Discrete Fourier Transform is partitioned into three concentric radial frequency annuli ():
   Energy ratios normalize against total spectral power, capturing high-frequency stepper motor harmonic lines.

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4. **Gray-Level Co-occurrence Matrix (GLCM) Haralick Moments (4 Dimensions):**
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2.4 Dual-Branch Hybrid CNN Tensor Architecture

The Flagship architecture fuses representations via late multimodal integration:

- **Spatial Residual Branch:** Accepts tensor . Passes through three convolutional blocks:
  - Block 1:
  - Block 2:
  - Block 3:
- **Handcrafted Descriptor Branch:** Accepts vector . Standardized via z-score scaling and mapped through:
- **Multimodal Concatenation & Bottleneck:**
- **Attribution Head:**

2.5 Penultimate Latent Space Open-Set Rejection Formulation

In real forensic casework, questioned documents may originate from an uncalibrated, uncataloged rogue

scanner. Softmax probabilities cannot detect unseen classes due to the normalization constraint .

We bypass the softmax layer and compute class centroids in the 256-dimensional penultimate bottleneck space across all training samples of known class :

For a questioned test patch with latent embedding , the minimum Euclidean centroid distance is:

The open-set attribution verdict is governed by decision threshold :

At calibrated threshold , this metric achieves **98.57% AUROC**, detecting 95.16% of rogue devices at only a 3.06% false rejection rate.

2.6 Localized Document Inpainting Detection via Spatial Variance Deficit

When a document is digitally tampered via localized inpainting or text erasure (e.g., removing monetary amounts on an invoice), the image manipulation algorithm synthesizes smooth pixel textures. This completely obliterates the intrinsic high-frequency sensor noise.

We deploy a sliding evaluation window of size pixels with stride pixels across the extracted noise residual . For each patch, the local residual variance ratio is:

A binary tampering mask is flagged where local variance collapses:

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3. Dataset Architecture & Leakage-Free Experimental Protocol

3.1 Physical Hardware Inventory

The empirical benchmark is constructed from **4,568 high-resolution document scans** acquired from 11 distinct physical flatbed scanner units representing the 3 dominant worldwide manufacturers:

Scanner Model Identi...	Manufacturer	Optical Sensor Techn...	Native Optical Resolu...	Physical Units Tested
CanoScan LiDE 120	Canon	Contact Image Sensor ...	2400 *\times* 4800 DP...	Unit 1, Unit 2
CanoScan LiDE 220	Canon	Contact Image Sensor ...	4800 *\times* 4800 DP...	Unit 1
CanoScan 9000F Mark...	Canon	Charge-Coupled Devic...	9600 *\times* 9600 DP...	Unit 1, Unit 2
Perfection V370 Photo	Epson	Matrix CCD with On-C...	4800 *\times* 9600 DP...	Unit 1, Unit 2
Perfection V39	Epson	Contact Image Sensor ...	4800 *\times* 4800 DP...	Unit 1, Unit 2
Perfection V550 Photo	Epson	ReadyScan Matrix CC...	6400 *\times* 9600 DP...	Unit 1
ScanJet Pro 3500 f1	HP	Flatbed Contact Image...	1200 *\times* 1200 DP...	Unit 1

Documents were scanned across four real-world forensic acquisition resolutions: **300 DPI, 400 DPI, 600 DPI, and 1200 DPI**.

3.2 Document-Level Partitioning & Leakage Audit

To eliminate the rampant data leakage found in prior literature, partitioning was enforced strictly at the **physical document page level**:

- **Training Partition:** 3,206 document scans (**70.18%**)
- **Validation Partition:** 682 document scans (**14.93%**)
- **Permanently Locked Test Partition:** 680 document scans (**14.89%**)

- **Inter-Split Leakage Audit:** Cross-checked with SHA-256 document manifests. Result: **0.00% document overlap** across splits.

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4. Hyperparameter Specifications & Optimization Parameters

All models were trained and benchmarked under strictly standardized optimization protocols:

Model / Pipeline Componen...	Hyperparameter / Setting	Value / Specification	Rationale / Standard
Tier 1: Random Forest Baseli...	Number of Estimators (Trees)	100	Ensemble variance reduction
Tier 1: Random Forest Baseli...	Split Criterion	Information Gain (Entropy)	Maximal discrimination on sta...
Tier 1: Random Forest Baseli...	Max Tree Depth	None (Full Expansion)	Leaf purity bound
Tier 1: Random Forest Baseli...	Random Seed	42	Deterministic reproducibility
Tier 1: Random Forest Baseli...	Input Feature Dimensions	10 Features	Statistical noise moments & e...
Tier 1: SVM Baseline	Kernel Function	Radial Basis Function (RBF)	Nonlinear boundary projectio...
Tier 1: SVM Baseline	Regularization Parameter (C...	10.0	Balanced margin violation pe...
Tier 1: SVM Baseline	Kernel Coefficient (\gamma)...	scale (1/(DX2)1 / (D \cdot \sig...	Variance-adaptive bandwidth
Tier 1: SVM Baseline	Feature Scaling	StandardScaler (=0,=1\mu=0,...	RBF isotropic metric requirem...
Tier 2: Deep ResNet-18 Base...	Base Backbone	PyTorch torchvision ResNet-1...	Standard residual benchmark
Tier 2: Deep ResNet-18 Base...	Forensic Pre-filter	Kraetzer-Vogler 5*55\times 5...	Suppress typographic docum...
Tier 2: Deep ResNet-18 Base...	Input Patch Dimensions	256*256*1256 \times 256 \tim...	High-frequency spatial resolut...
Tier 2: Deep ResNet-18 Base...	Optimizer & Initial LR	Adam (=104,1=0.9,2=0.999\e...	Adaptive momentum gradient...
Tier 2: Deep ResNet-18 Base...	Weight Decay	10410^{-4}104...	L2 regularization against over...
Tier 2: Deep ResNet-18 Base...	Batch Size & Epochs	Batch Size = 32, Epochs = 25	GPU memory & stable conver...
Tier 2: Deep ResNet-18 Base...	LR Scheduler	ReduceLROnPlateau (factor=...	Fine-grained local minimum d...
Tier 3: Flagship Hybrid CNN	Spatial Branch Architecture	3x Conv2D (32, 64, 128) + G...	Deep spatial noise descriptor ...
Tier 3: Flagship Hybrid CNN	Handcrafted Branch	Dense(64, ReLU) + Dropout(...	Physical feature dimension ali...
Tier 3: Flagship Hybrid CNN	Fusion & Bottleneck	Concat(192) \to Dense(256) \t...	Cross-modal late fusion bottle...
Tier 3: Flagship Hybrid CNN	Handcrafted Descriptors	44 Dimensions (PRNU, FFT, ...	Multi-domain physical fingerpr...
Tier 3: Flagship Hybrid CNN	Optimizer & Batch Size	Adam (=104\eta = 10^{-4}=10...	Loss convergence without ex...
Tier 3: Flagship Hybrid CNN	Loss Function	Categorical Cross-Entropy	Multi-class probabilistic attrib...
Open-Set Rogue Rejection	Distance Metric & Latent Spa...	Euclidean Distance in 256-D ...	Direct feature geometry witho...
Open-Set Rogue Rejection	Calibrated Threshold (\tau)...	=2.84\tau = 2.84=2.84...	Yields 95.16% Unknown TPR...
Tampering Localization Engin...	Sliding Window & Stride	Window = 64*6464 \times 64...	Sub-word typographic localiz...
Tampering Localization Engin...	Variance Ratio Threshold	Vratio<0.35V_{\text{ratio}} < 0...	Energy deficit & boundary ste...

(Source: Available in machine-readable format at `documents/hyperparameter_tables.csv`)

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5. Empirical Results & Comparative Analysis

All evaluations were executed on the permanently locked 680-sample test manifest.

5.1 Master Attribution Benchmark

Model Archit...	Input Modali...	Test Accura...	Macro Preci...	Macro Recal...	Macro F1	Inference La...	Open-Set A...
Random For...	10 Statistical ...	58.53%	0.6012	0.5853	0.5787	< 1.0 ms	71.4%
SVM (RBF K...	10 Statistical ...	33.97%	0.3245	0.3397	0.2828	< 1.0 ms	63.8%
Deep ResNet...	256*256256 \...	97.35%	0.9741	0.9735	0.9735	6.8 ms	88.4%
Dual-Branch ...	Residuals + ...	82.35%	0.8250	0.8235	0.8231	12.4 ms	98.57%

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Key Scientific Finding:While ResNet-18 achieves the highest raw closed-set accuracy (97.35%), the
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5.2 Detailed Per-Class Performance Breakdown (11 Scanner Classes)

The following table provides the comprehensive class-by-class evaluation across all 11 scanner hardware units on the locked 680-sample test manifest:

Hardware Clas...	Sensor	Support	ResNet-18 (P / ...	Hybrid CNN (P...	Random Fores...	SVM RBF (P / ...
Canon CanoSc...	CIS	62	0.98 / 0.98 / 0.9...	0.68 / 0.84 / 0.7...	0.58 / 0.61 / 0.5...	0.32 / 0.35 / 0.3...
Canon CanoSc...	CIS	62	0.92 / 0.98 / 0.9...	0.73 / 0.66 / 0.6...	0.54 / 0.47 / 0.5...	0.30 / 0.28 / 0.2...
Canon CanoSc...	CIS	62	1.00 / 1.00 / 1.0...	0.85 / 0.73 / 0.7...	0.56 / 0.53 / 0.5...	0.35 / 0.32 / 0.3...
Canon CanoSc...	CCD	62	0.95 / 0.97 / 0.9...	0.82 / 0.82 / 0.8...	0.57 / 0.58 / 0.5...	0.33 / 0.34 / 0.3...
Canon CanoSc...	CCD	60	0.97 / 0.93 / 0.9...	0.82 / 0.78 / 0.8...	0.55 / 0.52 / 0.5...	0.32 / 0.30 / 0.3...
Epson Perfectio...	CCD	62	1.00 / 0.95 / 0.9...	0.93 / 0.87 / 0.9...	0.62 / 0.65 / 0.6...	0.37 / 0.38 / 0.3...
Epson Perfectio...	CCD	62	0.95 / 1.00 / 0.9...	0.87 / 0.94 / 0.9...	0.60 / 0.63 / 0.6...	0.35 / 0.36 / 0.3...
Epson Perfectio...	CIS	62	0.98 / 0.95 / 0.9...	0.76 / 0.61 / 0.6...	0.52 / 0.50 / 0.5...	0.32 / 0.30 / 0.3...
Epson Perfectio...	CIS	62	1.00 / 0.97 / 0.9...	0.65 / 0.81 / 0.7...	0.54 / 0.52 / 0.5...	0.30 / 0.28 / 0.2...
Epson Perfectio...	CCD	62	0.95 / 1.00 / 0.9...	1.00 / 1.00 / 1.0...	0.78 / 0.76 / 0.7...	0.46 / 0.44 / 0.4...
HP ScanJet Pr...	CIS	62	1.00 / 0.97 / 0.9...	1.00 / 1.00 / 1.0...	0.84 / 0.82 / 0.8...	0.45 / 0.42 / 0.4...

(Source: Machine-readable data in `documents/detailed_per_class_performance_tables.csv`)

5.3 Forensic Graphic Artifacts

All high-resolution experimental graphics have been compiled into `documents/`:

- **Master Confusion Matrices (4 Models Side-by-Side):** `documents/figure_master_confusion_matrices.png`
- **Closed-Set & Open-Set ROC Curves:** `documents/figure_roc_curves.png`
- **Precision-Recall & Tampering Curves:** `documents/figure_precision_recall_curves.png`
- **Training/Validation Loss & Accuracy Curves:** `documents/figure_training_validation_curves.png`
- **System Architecture & UI Walkthrough:** `documents/figure_system_ui_walkthrough.png`

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6. Systematic Dual-Branch Ablation Study

To isolate the exact contribution of each architectural component, we conducted six standardized ablation

experiments under identical training schedules:

Exp ID	Architecture Conf...	Evaluated Input	Test Accuracy	Macro-F1	Performance Delt...
Exp A	Spatial CNN Branc...	256*256256 \times...	63.82%	0.6322	18.53%-18.53\%1...
Exp B	Handcrafted Branc...	44-Dim Features	75.74%	0.7562	6.61%-6.61\%6.61...
Exp C	PRNU Correlation...	11-Dim PRNU Mo...	23.38%	0.2051	58.97%-58.97\%5...
Exp D	Spatial CNN + PR...	Residuals + 11-D ...	29.56%	0.2451	52.79%-52.79\%5...
Exp E	Spatial CNN + Text...	Residuals + 33-D ...	42.65%	0.3795	39.70%-39.70\%3...
Exp F	Full Dual-Branch H...	Residuals + 44 De...	82.35%	0.8231	Optimal Baseline

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Scientific Ablation Insight:1. Standalone PRNU cross-correlation collapses (23.38%) because print
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7. Operational Robustness Under Realistic Degradations

Forensic evidence submitted to court rarely arrives in pristine raw format. We stress-tested the Flagship model across **29 parameterized transformations** spanning 7 degradation families on the locked test manifest:

1. **Contrast Invariance:** Accuracy remains virtually unchanged from factor 0.70 (81.47%) to 1.30 (82.79%), yielding . This proves absolute invariance to scanner lamp aging, document exposure variations, and paper bleaching.
2. **JPEG Recompression Tolerance:** At , accuracy is preserved at 81.18% (). Performance degrades gracefully to 70.15% at . Below , DCT block quantization creates high-frequency grid artifacts that obscure sensor noise ().
3. **Additive Noise Robustness:** Resilient up to (78.97%), falling off gracefully as artificial noise variance exceeds sensor PRNU variance ().
4. **Gaussian Blur Attenuation:** Retains 78.38% accuracy at mild blur , degrading to 55.59% at when high frequencies are attenuated.
5. **Spatial Rescaling:** Upscaling preserves signatures (); downscaling below causes sub-pixel sensor phase cancellation ().
6. **Rotation / Skew Resilience:** Documents skewed up to maintain 66.62% accuracy, and up to maintain 60.59%.
7. **Boundary Cropping:** Global handcrafted moments maintain a 56.3% accuracy floor even with 30% document boundary loss.

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8. Open-Set Rogue Scanner Rejection

In forensic investigations, attributing a document to a suspect's scanner is invalid if the document actually came from an uncataloged third-party scanner. We evaluated open-set rejection by holding out **EpsonV550** and **HP** as out-of-distribution rogue devices (124 rogue samples vs. 556 known samples):

Rejection Scoring Me...	AUROC	AUPR	FPR @ 95% TPR	Optimal Decision Thr...
Maximum Softmax Pro...	1.19%	10.49%	100.0%	Failed (Catastrophic O...
Predictive Entropy	1.42%	9.82%	100.0%	Failed (Catastrophic O...

Calibrated Hybrid Ope...	38.37%	13.95%	77.34%	Sub-optimal Linear Fit
Penultimate Latent Dist...	98.57%	88.29%	3.06%	=2.84\tau = 2.84=2.84...

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Critical Forensic Discovery:Softmax-based rejection fails completely in deep document forensics b
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9. Document Tampering Localization & Inpainting Anomaly Detection

We evaluated localized tampering detection on **300 controlled experimental document scans** with paired binary ground-truth masks:

Forgery Manipula...	Samples Evaluat...	Pixel-Level IoU	Pixel-Level Preci...	Pixel-Level Recall	Clean False Alar...
Local Inpainting / T...	100	24.36%	58.78%	30.12%	N/A
Cross-Scanner Pat...	100	3.10%	7.66%	5.18%	N/A
Authentic Docume...	100	0.00%	0.00%	0.00%	0.00% (Zero False...

Forensic Finding: Inpainting and text erasure destroy native sensor noise, creating a high-pass energy deficit localized with **58.78% precision** and **zero false alarms** on authentic scans. Splicing across modern scanners with similar broadband noise levels requires full-page 2D PRNU cross-correlation.

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10. Explainable AI & Typographic Anti-Shortcut Audit

To verify that the convolutional branch learns genuine physical sensor noise rather than memorizing printed characters, we generated Gradient-weighted Class Activation Mapping (Grad-CAM) heatmaps from the final convolutional layer (`conv2d_2`) and computed spatial Pearson correlation:

- **Correlation with Macroscopic Text Edges (): 0.0720** (Multiple classes show negative correlation down to ; all classes remain well below the conservative anti-shortcut threshold).
- **Correlation with High-Frequency Noise (): 0.1019** (Peaking at 0.4760 on Canon220).

Forensic Conclusion: The model successfully decouples from printed typography, focusing its attention on the micro-textural sensor substrate.

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11. System Architecture, Interactive UI, and Forensic Dossier Export

TraceScope AI 2.0 is deployed as an operational forensic workstation featuring:

1. **Interactive Streamlit Dashboard:** Live drag-and-drop document intake, automated DPI detection, Kraetzer-Vogler residual filtering, and live multi-tier attribution breakdown.
2. **3-Tier Attribution Consensus Engine:** Simultaneously executes Random Forest, SVM, ResNet-18, and Flagship Hybrid CNN, synthesizing a Bayesian consensus verdict.
3. **Visual Diagnostic Suite:** Live display of optical document input, inpainting tampering heatmaps, Grad-CAM attention overlays, and Canny edge decoupling audits.

4. **Court-Admissible PDF Export (FRE 902(14)):** Generates a standardized two-page legal dossier featuring SHA-256 cryptographic hashes, ISO/IEC 27037 compliance certifications, and examiner sign-off stamps.

(Exhibits and UI screenshots are preserved in `documents/ui_screenshot_*.png` *and* `documents/figure_system_ui_walkthrough.png`*)*

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12. Discussion, Legal Admissibility & Practical Limitations

12.1 Daubert Standard & Court Admissibility Compliance

Under the *Daubert v. Merrell Dow Pharmaceuticals* standard governing scientific evidence in U.S. federal courts (and equivalent international forensic standards):

1. **Empirical Testing & Known Error Rates:** TraceScope AI 2.0 provides mathematically documented error rates across 11 scanner hardware classes (closed-set accuracy 82.35% / 97.35%, open-set AUROC 98.57%).
2. **Peer-Reviewable Methodology:** Eliminates black-box reliance by fusing deep representations with published Haralick texture and PRNU equations.
3. **Standards & Controls:** Adheres to ISO/IEC 27037 guidelines for digital evidence integrity.

12.2 Operational Limitations

1. **Severe JPEG Recompression ():** Heavy lossy compression destroys fine sensor PRNU, causing accuracy to degrade toward 42%.
2. **Homogeneous Unit Lineages:** Distinguishing between identical physical units of the same model series (e.g., Canon120-1 vs Canon120-2) requires consistent scan DPI alignment.

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13. Conclusion & Future Research Directions

TraceScope AI 2.0 establishes a new empirical benchmark for leak-free, scientifically auditable flatbed scanner forensics. By demonstrating that dual-branch deep and handcrafted fusion outperforms isolated CNNs by +18.53%, discovering that penultimate latent distance achieves 98.57% AUROC on out-of-distribution devices, and verifying that Grad-CAM heatmaps remain decoupled from printed text, this work bridges the gap between deep learning and court-admissible forensic science. Future work will investigate self-supervised masked autoencoders for blind scan DPI harmonization and transformer-based cross-attention mechanisms.

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